Unit 3: Existence, Uniqueness, and Geometry

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb from reading a slope field, through the phase plane and the existence–uniqueness test, into fixed points, contraction mappings, and a full Picard iteration. Solutions follow in Part III.

1. For the slope field of $\dfrac{dy}{dx} = x + y$, find the slope at the point $(1, 2)$ and the isocline along which every segment is horizontal.
2. For $\dfrac{dy}{dx} = y^2 - x$, find the curve along which every field segment has slope 2.
3. For $\dfrac{dy}{dx} = -\dfrac{x}{y}$, find the slope of the segment at $(3, 4)$, and describe the shape of the integral curves.
4. For the system $x' = y$, $y' = -x$, locate the equilibrium and classify it using the eigenvalues of its coefficient matrix.
5. For the linear system $x' = 2x$, $y' = 3y$, write the eigenvalues of the coefficient matrix and classify the origin.
6. For $\dfrac{dy}{dx} = \dfrac{x}{y-2}$, determine where the existence and uniqueness theorem fails to guarantee a unique solution.
7. Show that the initial value problem $\dfrac{dy}{dx} = \sqrt{y}$, $y(0) = 0$ has more than one solution, and explain which hypothesis of the theorem fails.
8. Find every fixed point of $g(x) = x^2$ and classify each as attracting or repelling.
9. Show that $g(x) = \cos x$ is a contraction on $[0, 1]$, and state what the Banach fixed point theorem then concludes.
10. **(Capstone)** For $\dfrac{dy}{dx} = y$ with $y(0) = 1$, carry out Picard iteration from $\varphi_0 = 1$: compute φ_1, φ_2, φ_3, and identify the function the iterates converge to.

Part III — Complete Solutions

Solution 1. The slope is just f evaluated at the point.

$$f(1,2) = 1 + 2 = 3.$$

The horizontal segments lie on the isocline $f = 0$:

$$x + y = 0 \implies y = -x.$$

Solution 2. Set f equal to the target slope and solve.

$$y^2 - x = 2 \implies x = y^2 - 2.$$

Every segment crossing the parabola $x = y^2 - 2$ has slope 2.

Solution 3. Substitute, keeping the leading sign:

$$f(3,4) = -\frac{3}{4}.$$

The integral curves satisfy $y\,dy = -x\,dx$, so $\frac{y^2}{2} = -\frac{x^2}{2} + C_1$, that is

$$x^2 + y^2 = C \qquad \text{(concentric circles)}.$$

Solution 4. Equilibria need both derivatives zero:

$$y = 0 \text{ and } -x = 0 \implies (0,0).$$

The coefficient matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ has characteristic equation $\lambda^2 + 1 = 0$:

$$\lambda = \pm i \ \text{(purely imaginary)}.$$

Purely imaginary eigenvalues give a center: trajectories are closed orbits circling the origin.

Solution 5. The system is decoupled, so the coefficient matrix $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$ is diagonal and its eigenvalues sit on the diagonal:

$$\lambda_1 = 2, \quad \lambda_2 = 3.$$

Both are real and positive, so the origin is an unstable node (a source) — every nearby trajectory flows outward.

Solution 6. Locate where f and $\frac{\partial f}{\partial y}$ lose continuity. With $f = \dfrac{x}{y-2}$,

$$\frac{\partial f}{\partial y} = -\frac{x}{(y-2)^2},$$

and both blow up when the denominator vanishes:

$$y - 2 = 0 \implies y = 2.$$

The theorem fails to guarantee uniqueness along the line $y = 2$; everywhere else a unique solution exists.

Solution 7. Here $f = \sqrt{y}$ is continuous at $y = 0$, but

$$\frac{\partial f}{\partial y} = \frac{1}{2\sqrt{y}}$$

is *not* continuous at $y = 0$ — it blows up. So uniqueness is not guaranteed, and indeed two solutions pass through the origin:

$$y = 0 \qquad \text{and} \qquad y = \frac{x^2}{4} \quad (x \geq 0),$$

since $\frac{d}{dx}\left(\frac{x^2}{4}\right) = \frac{x}{2} = \sqrt{x^2/4}$. The failed hypothesis is continuity of $\frac{\partial f}{\partial y}$.

Solution 8. Solve $g(x) = x$:

$$x^2 = x \implies x(x-1) = 0 \implies x = 0,\ 1.$$

Classify with $|g'(x)| = |2x|$:

$$|g'(0)| = 0 < 1 \text{ (attracting)}, \qquad |g'(1)| = 2 > 1 \text{ (repelling)}.$$

Solution 9. A bound $|g'(x)| \leq k < 1$ on an interval makes g a contraction there. With $g(x) = \cos x$,

$$|g'(x)| = |\sin x|,$$

which increases on $[0, 1]$ and so is largest at $x = 1$:

$$|\sin x| \leq \sin 1 \approx 0.841 < 1.$$

Thus g is a contraction with $k = \sin 1$. By Banach, g has a unique fixed point in $[0, 1]$, and iterating cosine from *any* start converges to it.

Solution 10. Use $\varphi_{n+1}(x) = 1 + \int_0^x \varphi_n(t)\,dt$ from $\varphi_0 = 1$.

$$\varphi_1 = 1 + \int_0^x 1\,dt = 1 + x.$$

$$\varphi_2 = 1 + \int_0^x (1+t)\,dt = 1 + x + \frac{x^2}{2}.$$

$$\varphi_3 = 1 + \int_0^x \left(1 + t + \frac{t^2}{2}\right) dt = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}.$$

These are the partial sums of the exponential series, so

$$\varphi_n \to 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots = e^x,$$

the true solution of $y' = y$, $y(0) = 1$.

Unit 3 Key Takeaway

This unit reads differential equations as geometry and guarantees rather than formulas. A slope field turns $y' = f(x, y)$ into a picture its integral curves follow; the phase plane does the same for systems. The existence and uniqueness theorem (f and $\partial f/\partial y$ continuous) says exactly one curve threads each point, and Picard iteration via the Banach contraction principle is the machine that proves it.